

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: YELETI RISHITHA

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ROLE: IoT Sensor Monitoring and Real-Time Data Collection

DESIGN / DEVELOPMENT DECISION:

I contributed to the IoT and real-time monitoring component of the system. My role focused on collecting environmental information through sensors, including rainfall, water level, and other relevant disaster indicators. I helped define how sensor readings could be monitored and transferred to the disaster-management platform for further analysis.

CHALLENGE FACED AND SOLUTION:

The major challenge was handling continuously changing sensor information and ensuring that important changes could be detected quickly. I addressed this by focusing on real-time monitoring and threshold-based conditions that can identify abnormal readings. This allows the system to generate warnings when dangerous conditions are detected.

LEARNING GAINED:

I learned how Internet of Things devices can be integrated with software systems for real-time monitoring. I understood the importance of continuous data collection, sensor accuracy, data processing, and communication between hardware and software components.

SDG RELEVANCE:

The project supports SDG 11 by improving disaster preparedness and helping communities respond to environmental threats. Real-time monitoring can provide useful information before and during flooding situations.

COURSE OUTCOME CONTRIBUTION:

My contribution supports course outcomes related to system architecture, real-time data processing, requirements analysis, and software integration. I learned how hardware-generated information can become part of a complete software solution.

POSSIBLE IMPROVEMENT:

I would improve the IoT module by adding more sensors, wireless communication, automatic fault detection, battery monitoring, and improved connectivity for locations with limited network availability.